

Chapter 8:

Integration Techniques and Improper Integrals

Section 8.8: Improper Integrals

Infinite Discontinuity

A function f has an infinite discontinuity at $x = a$ if one of the following satisfied

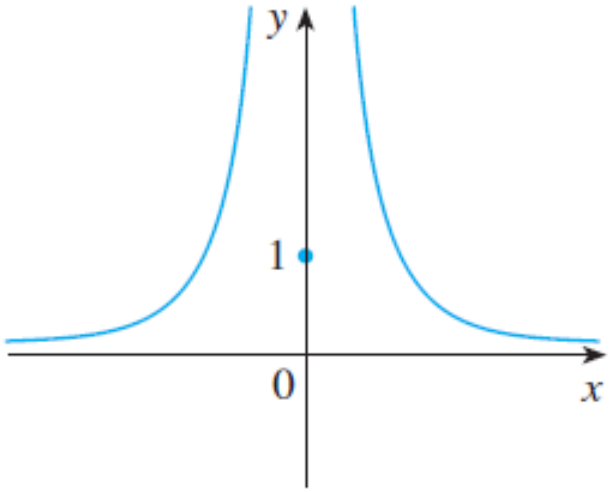
$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a} f(x) = -\infty \qquad \lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

In other words, if $x = a$ is a vertical asymptote of the curve $y = f(x)$, then f has an infinite discontinuity at $x = a$.

EXAMPLE

$$f(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$



Since $\lim_{x \to 0} f(x) = \lim_{x \to 0} \dfrac{1}{x^2} = \infty$, the function f has an infinite disconituity at $x = 0$.

So far, we considered definite integrals of the form $\int_a^b f(x)\ dx$ where f is a continuous function on a closed interval $[a, b]$ for which a and b are numbers.

Such definite integrals are called "**proper integrals**".

Improper integrals are definite integrals in which
(1) the interval of integration is infinite (that is, one or both limits of integration is (are) infinite) or
(2) the integrand has an infinite discontinuity on the interval $[a, b]$.

EXAMPLE

(1) $\int_0^\infty e^{-x}\ dx$ ← This is an improper integral because the interval of integration is infinite

(2) $\int_1^3 \frac{1}{x-1}\ dx$ ← This is an improper integral because the integrand has infinite discontinuity at $x = 1$.

(3) $\int_{-\infty}^3 \frac{1}{x-1}\ dx$ ← This is an improper integral for both reasons.

Type 1: Infinite Intervals

Consider the infinite region S that lies under the curve $y = 1/x^2$, above the x-axis, and to the right of the line $x = 1$. You might think that, since S is infinite in extent, its area must be infinite, but let's take a closer look. The area of the part of S that lies to the left of the line $x = t$ (shaded in Figure 1) is

$$A(t) = \int_1^t \frac{1}{x^2}\,dx = -\frac{1}{x}\Big]_1^t = 1 - \frac{1}{t}$$

Notice that $A(t) < 1$ no matter how large t is chosen.

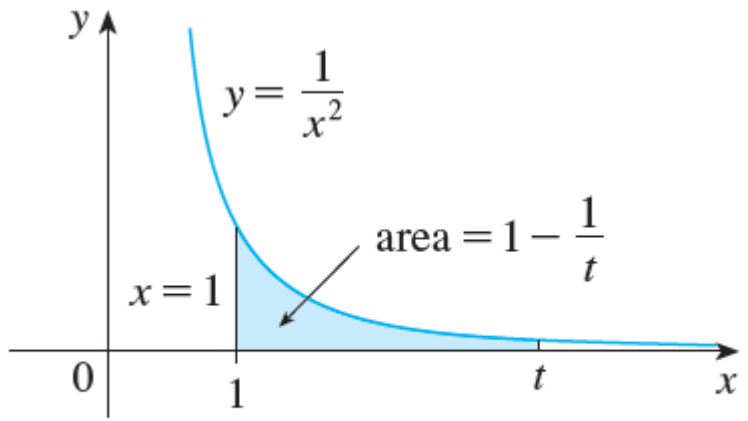
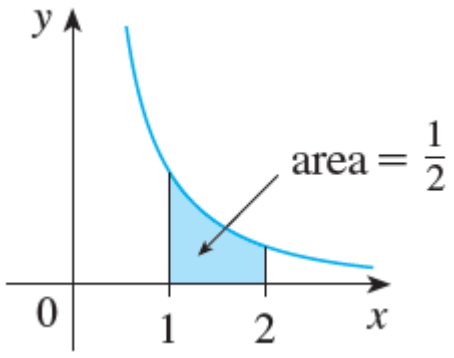
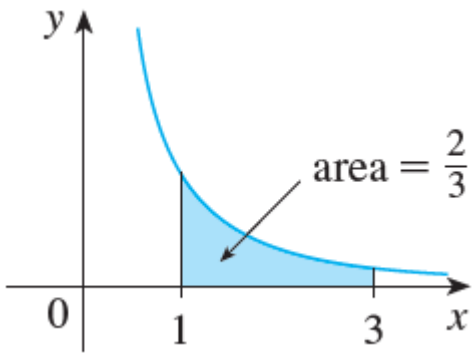


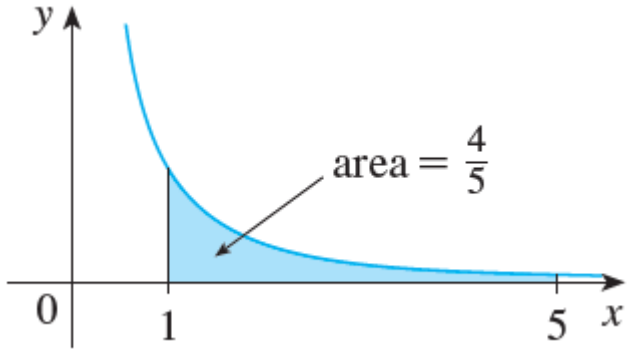
FIGURE 1

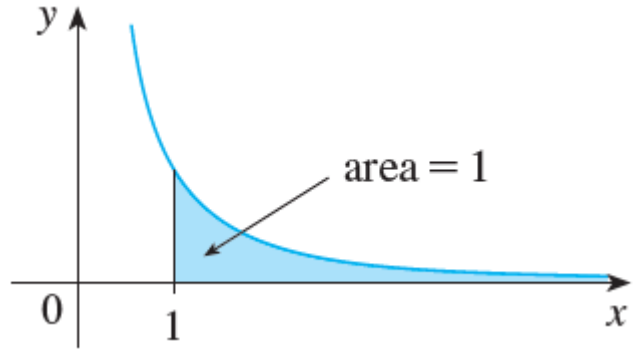
We also observe that

$$\lim_{t\to\infty} A(t) = \lim_{t\to\infty}\left(1 - \frac{1}{t}\right) = 1$$









Type 1: Infinite Intervals

Consider the infinite region S that lies under the curve $y = 1/x^2$, above the x-axis, and to the right of the line $x = 1$. You might think that, since S is infinite in extent, its area must be infinite, but let's take a closer look. The area of the part of S that lies to the left of the line $x = t$ (shaded in Figure 1) is

$$A(t) = \int_1^t \frac{1}{x^2}\,dx = -\frac{1}{x}\Big]_1^t = 1 - \frac{1}{t}$$

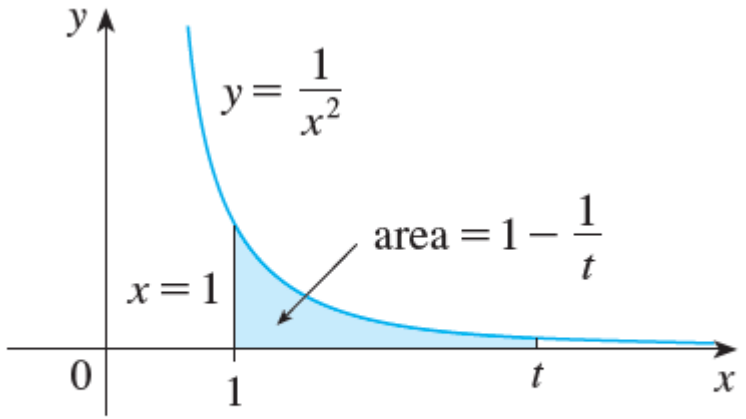


FIGURE 1

The area of the shaded region approaches 1 as $t \to \infty$ (see Figure 2), so we say that the area of the infinite region S is equal to 1 and we write

$$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{t\to\infty} \int_1^t \frac{1}{x^2}\,dx = 1$$

1 Definition of an Improper Integral of Type 1

(a) If $\int_a^t f(x)\,dx$ exists for every number $t \geqslant a$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^t f(x)\,dx$$

provided this limit exists (as a finite number).

(b) If $\int_t^b f(x)\,dx$ exists for every number $t \leqslant b$, then

$$\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty} \int_t^b f(x)\,dx$$

provided this limit exists (as a finite number).

The improper integrals $\int_a^{\infty} f(x)\,dx$ and $\int_{-\infty}^b f(x)\,dx$ are called **convergent** if the corresponding limit exists and **divergent** if the limit does not exist.

(c) If both $\int_a^{\infty} f(x)\,dx$ and $\int_{-\infty}^a f(x)\,dx$ are convergent, then we define

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^a f(x)\,dx + \int_a^{\infty} f(x)\,dx$$

In part (c) any real number a can be used (see Exercise 107).

EXAMPLE

Determine whether the integral converges or diverges. $\int_{-\infty}^{0} x\, e^{x}\, dx.$

SOLUTION

The integral converges and its value is $I = -1$

EXAMPLE

Determine whether the integral converges or diverges. $\int_0^{\infty} \frac{2x}{1+x^2}\, dx.$

SOLUTION

The integral diverges

EXAMPLE

Determine whether the integral converges or diverges. $\int_{-\infty}^{\infty} \frac{1}{1+x^2}\, dx.$

SOLUTION

The integral converges and its value is $I = \pi$

EXAMPLE 4 For what values of p is the integral

$$\int_1^\infty \frac{1}{x^p}\,dx$$

convergent?

SOLUTION

$$\int_1^\infty \frac{1}{x^p}\,dx = \lim_{t\to\infty}\int_1^t x^{-p}\,dx = \lim_{t\to\infty}\left.\frac{x^{-p+1}}{-p+1}\right]_{x=1}^{x=t} = \lim_{t\to\infty}\frac{1}{1-p}\left[\frac{1}{t^{p-1}} - 1\right]$$

If $p > 1$, then $p - 1 > 0$, so as $t \to \infty$, $t^{p-1} \to \infty$ and $1/t^{p-1} \to 0$. Therefore

$$\int_1^\infty \frac{1}{x^p}\,dx = \frac{1}{p-1} \qquad \text{if } p > 1$$

and so the integral converges. But if $p < 1$, then $p - 1 < 0$ and so

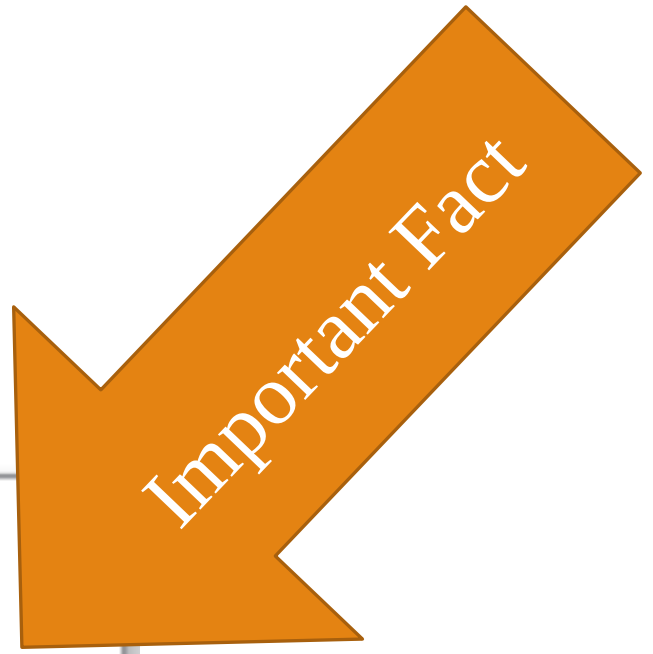
$$\frac{1}{t^{p-1}} = t^{1-p} \to \infty \qquad \text{as } t \to \infty$$

and the integral diverges.

EXAMPLE 4 For what values of p is the integral

$$\int_1^\infty \frac{1}{x^p}\,dx$$

convergent?



THEOREM 8.7 A Special Type of Improper Integral

$$\int_1^\infty \frac{dx}{x^p} = \begin{cases} \dfrac{1}{p-1}, & p > 1 \\ \text{diverges}, & p \le 1 \end{cases}$$

EXAMPLE

Determine whether the integral converges or diverges. $\int_1^\infty \frac{1}{x^{4.9}}\,dx.$

SOLUTION

EXAMPLE

Determine whether the integral converges or diverges. $\int_1^\infty \frac{1}{\sqrt{x}}\,dx.$

SOLUTION

EXAMPLE

Determine whether the integral converges or diverges. $\int_{0.5}^\infty \frac{1}{x^2}\,dx.$

SOLUTION

Type 2: Discontinuous Integrands

Suppose that f is a positive continuous function defined on a finite interval $[a, b)$ but has a vertical asymptote at b. Let S be the unbounded region under the graph of f and above the x-axis between a and b. (For Type 1 integrals, the regions extended indefinitely in a horizontal direction. Here the region is infinite in a vertical direction.) The area of the part of S between a and t (the shaded region in Figure 7) is

$$A(t) = \int_a^t f(x)\,dx$$

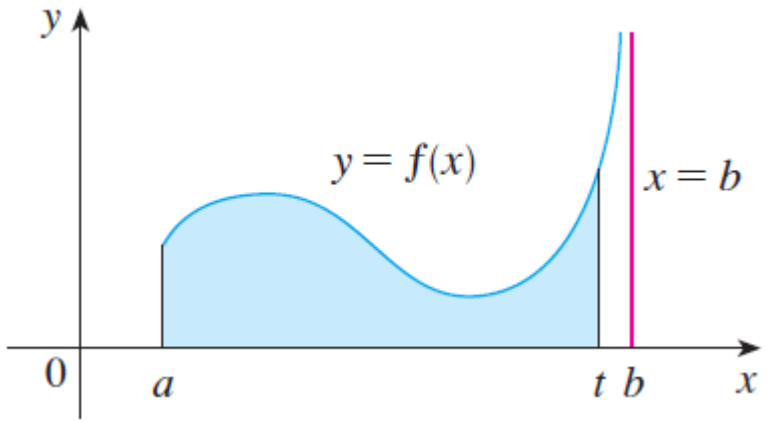


FIGURE 7

If it happens that $A(t)$ approaches a definite number A as $t \to b^-$, then we say that the area of the region S is A and we write

$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx$$

3 Definition of an Improper Integral of Type 2

(a) If f is continuous on $[a, b)$ and is discontinuous at b, then

$$\int_a^b f(x)\,dx = \lim_{t \to b^-} \int_a^t f(x)\,dx$$

if this limit exists (as a finite number).

(b) If f is continuous on $(a, b]$ and is discontinuous at a, then

$$\int_a^b f(x)\,dx = \lim_{t \to a^+} \int_t^b f(x)\,dx$$

if this limit exists (as a finite number).

The improper integral $\int_a^b f(x)\,dx$ is called **convergent** if the corresponding limit exists and **divergent** if the limit does not exist.

(c) If f has a discontinuity at c, where $a < c < b$, and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent, then we define

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Parts (b) and (c) of Definition 3 are illustrated in Figures 8 and 9 for the case where $f(x) \geqslant 0$ and f has vertical asymptotes at a and c, respectively.

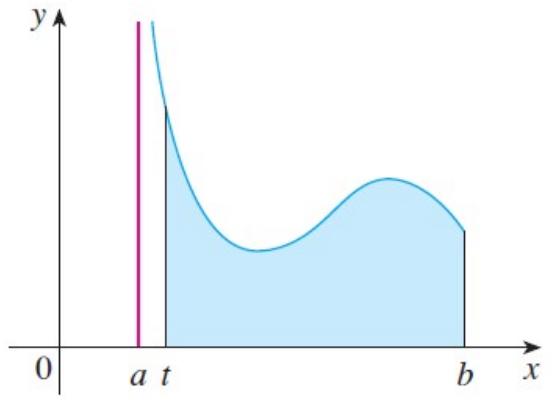


FIGURE 8

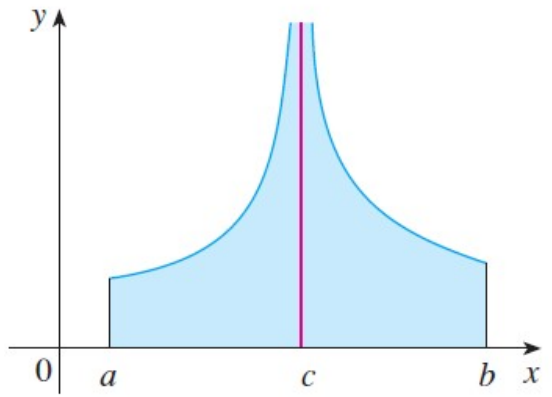


FIGURE 9

EXAMPLE

Determine whether the integral converges or diverges. $\int_0^1 \frac{dx}{(x-1)^{\frac{2}{3}}}$.

SOLUTION

The integral converges and its value is $I = 3$.

EXAMPLE

Determine whether the integral converges or diverges. $\int_{-1}^{1} \frac{dx}{x+1}$.

SOLUTION

The integral diverges

EXAMPLE

Determine whether the integral converges or diverges. $\int_0^3 \frac{dx}{x-1}$.

SOLUTION

The integral diverges

EXAMPLE

Determine whether the integral converges or diverges. $\int_0^1 \ln x \ dx.$

SOLUTION

The integral converges and its value is $I = -1$.

EXAMPLE A Doubly Improper Integral

Evaluate $\int_0^\infty \frac{dx}{\sqrt{x}(x+1)}$.

SOLUTION **Read Example 9**

The integral converges and its value is $I = \pi$.

Solve exercises of section 8.8.

Start with the suggested problems mentioned in the syllabus.